# Learning Task 4 - Selection to the extreme: IIR filters

The purpose of this Learning Task is to let you practice with IIR filter design. This task will be much shorter than the first three learning tasks, but it reflects an important aspect of filter design: small transition bands and controlled behaviour in pass- and stopbands. One of the major advantages of IIR filters is their sharp transition between stop and passband with a relatively low number of coefficients.

**Q** Suppose we want to design an IIR bandpass filter passing the $[0.2\pi, 0.4\pi]$ band and removing all the other frequencies. We want at most 3dB ripple in the passband and at least 90dB attenuation in the stopband. The filter will be part of a resource limited embedded device, so we want to explore the effect of the budget of

a) $2 \times 8$

b) $2 \times 10$

c) $2 \times 12$

coefficients on the size of the largest possible stopband. Compare your filters to those of Learning Task 3.
**H** Search?
**A** We have

a) $2\times$ 8 $\rightsquigarrow$ stopband $[0, 0.1631\pi] \cup [0.4369\pi, \pi]$

b) $2\times$ 10 $\rightsquigarrow$ stopband $[0, 0.1852\pi] \cup [0.4148\pi, \pi]$

c) $2\times$ 12 $\rightsquigarrow$ stopband $[0, 0.1938\pi] \cup [0.4062\pi, \pi]$

The frequency response is shown in Figure 1. This calculation shows the low number of coefficients an IIR filter needs to meet a sharp transition band. A FIR filter would need much more coefficients. However, do notice its strong non-linear phase behaviour.

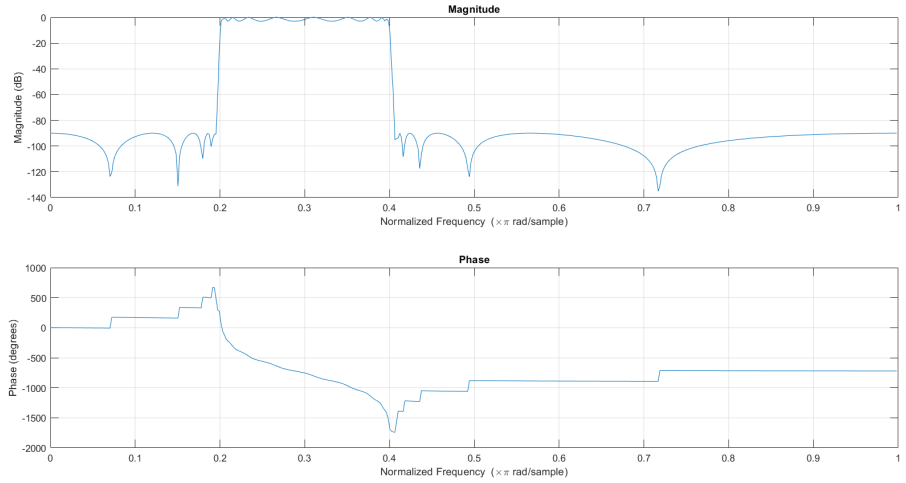



Figure 1: Frequency response of the filter with $2 \times 12$ coefficients.